INCH-POUND

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w/AMENDMENT 2
2 September 2025
SUPERSEDING
MIL-DTL-6193E
w/AMDENDMENT 1
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DETAIL SPECIFICATION

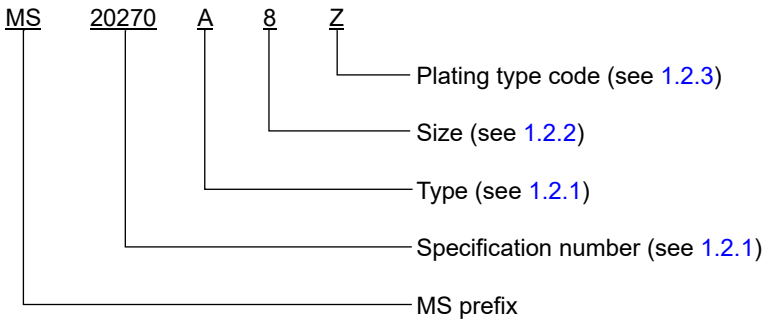
JOINTS, UNIVERSAL, PLAIN, LIGHT AND HEAVY DUTY, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers the requirements for joints, universal, plain, light and heavy duty, for use in intermittent operations (see 6.1).

1.2 Part or Identifying Number (PIN). The PIN consists of the letters MS (see 3.6), the basic specification number, a letter for type, a number for size, and a letter for plating type.



1.2.1 Classification. The universal joints are of the following classes and types:

Class 1 – Light duty specification sheet MS20270.

Type A – Square end.
Type B – Round end.

Class 2- Heavy duty specification sheet MS20271.

Type A – Not applicable.
Type B – Round end.

Comments, suggestions, or questions on this document should be addressed to: Commander, DLA Land and Maritime, Attn: VAI, P.O. Box 3990, Columbus, Ohio, 43218-3990 or email: FluidFlow@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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1.2.2 Size. See Table I and the individual specification sheets, specified in 1.2.1.

TABLE I. Class to size designator.

Class (see 1.2.1)	Size designator (Nominal size of joint)	Nominal OD of universal joint (inch)	Nominal size of universal joint (mm)
1	6 1/	0.375	9.53
1	8	0.500	12.70
1	10	0.625	15.88
1	12	0.750	19.05
1	14 1/	0.875	22.23
1	16	1.000	25.40
1	20	1.250	31.75
1	24	1.500	38.10
2	6	0.375	9.53
2	8	0.500	12.70
2	10	0.625	15.88
2	12	0.750	19.05
2	14	0.875	22.23
2	16	1.000	25.40
2	20	1.250	31.75
2	24	1.500	38.10

1/ Type B only.

1.2.3 Plating type codes.

Blank – Cadmium
P – Zinc phosphate
Z – Zinc
ZN – Zinc-Nickel Alloy

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include document cited in other sections of this specification, or recommended for additional information, or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or no they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks forma part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-16232 - Phosphate Coating, Heavy, Manganese or Zinc Base
MIL-PRF-7870 - Lubricating Oil, General Purpose, Low Temperature (NATO O-142)
MIL-PRF-32014 - Grease, Aircraft and Instrument
MIL-PRF-81322 - Grease, Aircraft, General Purpose, Wide Temperature Range, NATO CODE G-395

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130	-	Identification Marking of U.S. Military Property
MS20270	-	Joint – Universal, Plain, Light Duty
MS20271	-	Joint – Universal, Plain, Heavy Duty

(Copies of these documents are available online at https://quicksearch.dla.mil)

2.3 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

ASTM INTERNATIONAL

ASTM B633	-	Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel
ASTM B659	-	Standard Guide for Measuring Thickness of Metallic and Inorganic Coatings
ASTM D1149	-	Standard Test Methods for Rubber Deterioration – Cracking in an Ozone Controlled Environment

(Copies of these documents are available online at https://www.astm.org)

INTERNATION ORGANIZATION FOR STANDARDIZATION (ISO)

ISO/IEC 17025	-	General requirements for the Competence of Testing and Calibration Laboratories

(Copies of these documents are available online at https://www.iso.ch)

SAE INTERNATIONAL

SAE-AMS2417	-	Plating, Zinc-Nickel Alloy
SAE-AMS-M-7866	-	Molybdenum Disulfide, Technical, Lubrication Grade
SAE-AMS-QQ-P-416	-	Plating, Cadmium (Electro-deposited)

(Copies of these documents are available online at https://www.sae.org)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulation unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 Qualification. Universal joints furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable Qualified Products List (QPL) before contract award (see 4.3 and 6.3).

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3.3 Materials. Materials shall be as identified herein or as approved by the qualifying activity. However, when a definite material is not specified, a material shall be used which will enable the universal joint to meet the requirements of this specification. Acceptance of approval of any constituent material shall not be construed as a guarantee of acceptance of the finished product.

3.3.1 Recycled, recovered, or environmentally preferable, or biobased materials. Recycled, recovered, or environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements and promotes economically advantageous life cycle costs.

3.4 Design and construction.

3.4.1 Dimensions. Dimensions shall be as specified in MS20270 and MS20271.

3.4.2 Assembly. The joints shall be assembled with pins and bearings pinned or locked into place to prevent disassembly of the finished joint.

3.4.3 Hardness. The hub sections of joints (defined by area bounded by dimension H in MS20270 and MS20271) shall not exceed Rockwell hardness C-40.

3.4.4 Plating.

3.4.4.1 Steel parts (excluding corrosion resistant steel CRES). Plating types for steel parts, excluding corrosion resistant steel (CRES), shall be as described in Table II.

TABLE II. Plating types and standards.

Plating Code (see 1.2)	Plating Type	Applicable Specification
Blank	Cadmium	SAE-AMS-QQ-P, Type II, Class 2
P	Zinc-phosphate	MIL-DTL-16232, Type Z
Z	Zinc	ASTM B633
ZN	Zinc-nickel	SAE-AMS2417, Type 2, Grade B 1/

1/ The zinc-nickel alloy plate shall contain 12% to 16% nickel. The coating thickness shall be 315 µ-inches (8 µm) minimum.

3.4.4.2 Zinc and zinc phosphate. Not for use in aircraft and requires approval from the Program Office for all applications.

3.4.4.3 Verification of plating thickness. Verification of plating thickness shall be in accordance with ASTM B659.

3.4.5 Lubrication retaining covers (class 2). Lubrication retaining covers shall be made so that they exclude dirt and dust from the bearing surfaces, and shall retain the lubricant on those surfaces at all temperatures between -55°C and +71°C during operation.

3.4.5.1 Ozone resistance. The lubrication retaining cover when tested in accordance with 4.7.9 shall not exhibit any evidence of cracking.

3.4.5.2 Hazardous substances and ozone depleting chemicals. The ozone resistance test (see 4.7.9) may contain hazardous chemicals. Federal regulations and guidelines shall be followed to perform this test. For further information about toxic chemicals and the hazardous materials list, consult the Environmental Protection Agency web database at https://www.epa.gov/ozone-pollution.

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3.4.5.3 Without lubrication retaining covers. Class 1 universal joints shall be lubricated with oil conforming to MIL-PRF-7870.

3.4.5.4 With lubrication retaining covers. Class 2 universal joints shall be lubricated with a lubricant consisting of a uniform mixture of grease conforming to MIL-PRF-32014 or MIL-PRF-81322 with a 6 ± 0.5% by weight additive of molybdenum disulfide lubricant conforming to SAE-AMS-M-7866.

3.5 Performance.

3.5.1 Angularity. The joints shall be such a design and construction as to be operable at an angle of not less than 30° when measured between the axis of the hubs.

TABLE III. Performance.

Class (see 1.2.1)	Nominal size of universal joint (in.)	Torsional play		Axial tension and compression load	Static torque load ±2% (lb. in.)	Dynamic torque load ±2% (lb. in.)	Length of run endurance test hours
		Torque ±2% (lb. in.)	Limit ±2% (degrees)	±2% (lb.)			
1	0.375	4	1.00	200	175	26	2
1	0.500	4	0.80	200	250	38	2
1	0.625	4	0.64	300	500	75	2
1	0.750	4	0.53	400	1000	150	2
1	0.875	8	0.46	500	1750	262	2
1	1.000	8	0.40	600	2500	375	2
1	1.250	8	0.32	800	5000	750	2
1	1.500	8	0.27	1100	7500	1125	2
2	0.375	4	0.83	500	200	30	5
2	0.500	4	0.62	1000	600	90	5
2	0.625	4	0.50	1500	1080	162	5
2	0.750	4	0.42	2000	1900	285	5
2	0.875	8	0.36	3500	3000	450	5
2	1.000	8	0.32	5000	4700	705	5
2	1.250	8	0.24	7000	9500	1425	5
2	1.500	8	0.20	9000	14500	2175	5

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TABLE III. Performance – Continued. 1/

Class (see 1.2.1)	Nominal size of universal joint (mm)	Torsional play		Axial tension and compression load	Static torque load ±2% (N-m)	Dynamic torque load ±2% (N-m)	Length of run endurance test hours
		Torque ±2% (N-m)	Limit ±2% (degrees)	±2% (kg)			
1	9.53	0.45	1.00	90.72	19.77	2.94	2
1	12.70	0.45	0.80	90.72	28.25	4.29	2
1	15.88	0.45	0.64	136.08	56.49	8.47	2
1	19.05	0.45	0.53	181.44	112.98	16.95	2
1	22.23	0.90	0.46	226.80	197.72	29.60	2
1	25.40	0.90	0.40	272.16	282.46	42.37	2
1	31.75	0.90	0.32	362.87	564.92	84.74	2
1	38.10	0.90	0.27	498.95	847.39	127.11	2
2	9.53	0.45	0.83	226.80	22.60	3.39	5
2	12.70	0.45	0.62	453.59	67.79	10.17	5
2	15.88	0.45	0.50	680.39	122.02	18.30	5
2	19.05	0.45	0.42	907.18	214.67	32.20	5
2	22.23	0.90	0.36	1587.57	338.95	50.84	5
2	25.40	0.90	0.32	2267.96	531.03	79.65	5
2	31.75	0.90	0.24	2175.15	1073.36	161.00	5
2	38.10	0.90	0.20	4082.33	1638.28	245.74	5

1/ Metric equivalents are for information only

3.5.2 End and side play. When tested as specified in 4.7.2, the play in the joint shall not exceed the maximum allowable end and side play in Table IV.

TABLE IV. End and side play limits.

Class of universal joint	End play In. (mm)	Side play In. (mm)
Class 1	0.0035 (0.089)	0.0080 (0.020)
Class 2	0.0027 (0.069)	0.0055 (0.140)

3.5.3 Axial load. When tested as specified in 4.7.3, the universal joint shall support the axial tension and compression load shown in Table III for 30 to 31 seconds with angle “A” (see Figure 1) equal to 0°, without any permanent deformation or excessive tightness after the load is removed.

3.5.4 Tightness. When tested as specified in 4.7.3, the moment of force required to move the universal joint throughout the minimum angle of 30° shall not exceed 0.20 pound-inch ±2% (22.6 mN-m).

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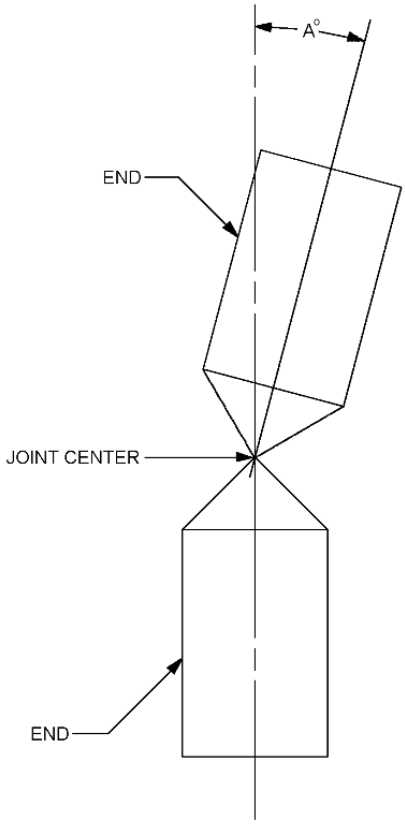


FIGURE 1. Universal joint.

3.5.5 Static torque load. Specimens shall be subjected to torque load conditions specified in 4.7.5.

3.5.6 Torsional play. The torsional play shall not exceed the applicable limit specified in Table III when tested as specified in 4.7.6.

3.5.7 Endurance test. The universal joint shall be endurance tested as specified in 4.7.7. Failure of the universal joint is indicated by the following:

a. 5° or more torsional play.
b. When lubrication retaining covers are in place:
    (1) Failure of the lubricant retaining cover.
    (2) Lubricant shall be retained by the lubrication retaining covers throughout the endurance run.

3.5.8 Lubrication retention (Class 2). The lubrication retention cover shall retain the lubricant and not crack or show signs of failure when tested as specified in 4.7.8. Slight bleeding of lubricant on surface of cover is permissible.

3.6 Identification of product. Each joint shall be permanently and legibly marked in accordance with MIL-STD-130. The minimum marking requirements shall be the PIN (see 1.2) and manufacturer's CAGE.

3.6.1 Use of MS or MIL designations. MS or MIL designations shall not be applied to a product, except for qualification test samples, nor referred to in correspondence or sales matter, until notification has been received from the qualifying activity that the product has been approved for military use.

3.7 Workmanship. All details of manufacture shall be in accordance with the best practices for high quality military parts. All components shall be uniform in quality and shall be free from irregularities or defects that could adversely affect performance or durability.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.3).
b. Conformance inspection (see 4.4).

4.2 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified herein.

4.2.1 Verification testing. The following tests and test methods assure universal joint integrity within typical operating conditions and applications. Alternate methods are allowed with prior approval by the qualifying activity. The test methods described herein are the preferred methods, and shall take precedence when alternate test methods result in differing or conflicting results.

4.2.2 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality, and quantity to permit performance of the required inspection shall be established and maintained by the contractor. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with ISO/IEC 17025.

4.2.3 Responsibility for compliance. All items shall meet all requirements of section 3 and 4. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.3 Qualification inspection. Qualification inspection shall consist of the examinations and test specified in Table V. Samples subjected to Group I testing shall be divided and subjected to Group II and Group III tests. Qualification inspections and tests shall be performed at a laboratory acceptable to the Government (see 6.3) on sample units produced with equipment and procedures normally used in production. Use of alternate materials, plating, or processes shall be identified in the product test document (see 3.3).

4.3.1 Qualification test articles. Samples for qualification shall be representative of the products proposed to be furnished to the Government. Qualification test articles shall consist of 10 universal joints for each type and size of universal joint for which qualification is desired.

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TABLE V. Qualification inspection and verification of qualification inspection.

Inspection	Requirement paragraph	Test method paragraph
Group I (10 samples) 1/		
Visual and mechanical inspection	3.4.1, 3.4.2, 3.6, 3.7	4.7.1
Verification of plating thickness 2/	3.4.4.3	---
Ozone resistance 2/	3.4.5.1	4.7.9
Hardness 3/	3.4.3	---
Group II (4 samples)		
Torsional play	3.5.6	4.7.6
End and side play	3.5.2	4.7.2
Tightness	3.5.4	4.7.4
Axial load	3.5.3	4.7.3
Tightness	3.5.4	4.7.4
Static torque load	3.5.5	4.7.5
Group III (4 samples)		
Endurance (sequence shown below)		
Endurance	3.5.7	4.7.7
Lubrication retention 4/	3.5.8	4.7.8
Torsional play	3.5.6	4.7.6

1/ Any samples not tested shall be retained as a reference standard.
2/ Applicable to Class 2 universal joints only. At the discretion of the qualifying activity, the manufacturer may verify conformance to 3.4.4.3 or 3.4.5.1 as applicable with a certificate of compliance stating testing has been completed in accordance with ASTM B659 or 4.7.9 as applicable or that material supplier, designs, and manufacturing processes have not changed.
3/ Hardness can be tested after all the other tests, on one sample for each type and size of universal joint for which qualification is desired.
4/ Three samples from endurance test with lubrication retention covers in place.

4.3.2 Failures. One or more failures shall be cause for refusal to grant qualification approval.

4.3.3 Test plans and qualification reports.

4.3.3.1 Test plans. Test plans shall be prepared and submitted in accordance with the requirements of the qualification activity. The method of qualification proposed by the contractor is subject to the approval of the qualification activity. Manufacturers shall discuss with the qualifying activity the test specimens and test plans. These plans should state specifically the component characteristic requirements to be monitored during the test, such as test fixtures, setup, conditions, and identification of the previous failure criteria shall be included as appropriate.

4.3.3.2 Qualification reports. Qualification reports shall be submitted in accordance with the requirements of the qualifying activity. As a minimum, manufacturers shall submit a report identifying test specimens and test results.

4.4 Conformance inspection. Conformance inspection shall consist of Group A (see 4.4.2) and Group B (see 4.5.1) inspections.

4.4.1 Inspection lot. An inspection lot shall consist of all universal joints produced under essentially the same conditions, and offered for inspection at one time.

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4.4.2 Group A inspection. Group A inspection shall be performed on a production lot basis and shall consist of the inspections specified in Table VI.

4.4.2.1 Sampling plan (Group A). Samples shall be selected in accordance with Table VII.

4.4.2.2 Failures (Group A). If one or more sample units fail to pass Group A inspection, the lot shall be screened for that particular defect, and defective parts shall be removed. A new sample of parts shall be selected in accordance with Table VII and all Group A tests again performed. If one or more defects are found in the second sample, the lot shall be rejected and shall not be supplied to this specification (see 4.5.4.1).

TABLE VI. Group A inspection.

Inspection	Requirement paragraph	Test method paragraph
Visual and mechanical inspection	3.4.1, 3.4.2, 3.6, 3.7	4.7.1
Verification of plating thickness	3.4.4.3	---
Hardness	3.4.3	---
Torsional play End and side play Tightness	3.5.6 3.5.2 3.5.4	4.7.6 4.7.2 4.7.4

TABLE VII. Inspection sample.

Production lot size	Accept on zero sample size
Under 200	3
Over 200	4

4.5 Periodic inspection (Group B). Periodic inspection shall consist of Group B inspections (see 4.5.1), except where the results of these inspections show noncompliance with the applicable requirements (see 4.5.4). Delivery of products which have passed Group A inspection shall not be delayed pending the results of these inspections.

4.5.1 Group B inspection. Group B inspection shall consist of the inspections specified in Table VIII in the order shown. Group B inspection shall be made on the lot of covers used in the manufacturing of universal joints, that have passed Group A inspections. Group B inspections shall be performed every 36 months.

TABLE VIII. Group B inspection.

Inspection	Requirement paragraph	Test method paragraph
Ozone resistance 1/	3.4.5.1	4.7.9

1/ At the discretion of the qualifying activity, the manufacturer may verify conformance to 3.4.5.1 as applicable with a certificate of compliance stating testing has been completed in accordance with ASTM D1149 or 4.7.9 as applicable. Applicable to Class 2 joints only.

4.5.1.1 Sampling plan. Ozone resistance samples may be randomly selected from sheets of finished manufactured material that will be used in the manufacturing of the universal joints.

4.5.2 Failures. If one or more sample units fail to pass Group B inspection, the samples shall be considered to have failed (see 4.5.4.1).

4.5.3 Disposition of sample units. Cover samples which have been subjected to Group B inspection, shall not be used on universal joints that are to be delivered on a contract or purchase order.

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4.5.4 Nonconformance.

4.5.4.1 Failures. If a sample fails to pass any Group A inspections (see Table VI) or Group B inspections (see Table VIII), the manufacturer shall immediately notify the qualifying activity and cognizant inspection activity of such failure. The manufacturer shall take corrective action on the materials, processes, or both as warranted, on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials and processes, and which are considered subject to the same failure.

4.5.4.2 Acceptance and shipment. Acceptance and shipment of the product shall be discontinued until corrective action acceptable to the qualifying activity has been taken. After the corrective action has been taken, Group A testing shall be repeated on additional samples. At the secretion of the qualifying activity, this may include all inspections or the inspection which the original sample failed. Individual, sampling, and periodic inspections, if applicable, may be reinstituted. However, final acceptance of the universal joints shall be withheld until the Group A inspection has shown that the corrective action was successful.

4.6 Additional QPL test and reporting requirements.

4.6.1 Retention of qualification. To retain qualification, the contractor shall submit a test report to the qualifying activity at 24-month intervals, and a report for Group B testing performed at 36-month intervals. The qualifying activity shall establish the initial reporting date. As a minimum, the report shall include the following:

a. Number of lots produced and tested, including lot and sample sizes for each lot.
b. Identify which tests were performed.
c. Quantities passed.
d. Quantities failed.
e. All reworked sampling lots shall be accounted for and identified. A summary of corrective action taken shall be included

4.6.2 Loss of product qualification.

4.6.2.1 Failure to meet test requirements. The manufacturer shall immediately notify the qualifying activity during the 24 or 36 month, as applicable, reporting period when the qualified product fails to meet the test and inspection requirements of this specification. The manufacturer shall identify and indicate what corrective action will be taken to correct the problem. Failure to take corrective action acceptable to the qualifying activity may result in removal of the product from the QPL.

4.6.2.2 Failure to submit summary test data report. Failure to submit a test report within 30 days after the end of the 24 or 36 month reporting period may result in removal of qualification for the product.

4.6.2.3 Change to manufacturing process, materials, or equipment. The manufacturer shall notify the qualifying activity, in writing, of any changes in the manufacturing process, materials, or equipment used to manufacture a QPL product. Subsequently, the qualifying activity will notify the manufacturer, in writing, if a full re-qualification, partial re-qualification, or if no additional testing is required as a result of these changes.

4.6.2.4 No production during reporting period. When no production occurs during the reporting period, a report shall be submitted to the qualifying activity certifying that the manufacturer still has the capability and facilities necessary to produce the QPL product. If there has been no production for 36 months, the manufacturer may be required, at the discretion of the qualifying activity, to submit QPL products to a full qualification inspection in accordance with this specification.

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4.7 Performance.

4.7.1 Examination of product. The universal joint and its manufacturing records shall be examined to verify conformance with the materials, construction, dimensions, identification requirements, and the applicable specification sheets.

4.7.2 End and side play (see 3.5.2). Both the end play and side play of the universal joints shall be measured and shall meet the requirements of 3.5.2. The following details shall apply:

a. For Class 2 universal joints, the lubricant retaining cover shall be removed for this test.
b. Lubricant shall be removed for this test.
c. For each measurement, a five-pound (2.27 kg) load shall be applied to obtain an initial reading.
d. The load shall then be reversed and a second reading taken.
e. The total play is defined as the play recorded between these two readings.
f. The side play of the universal joint shall be measured in the same plane as an inspection hole.
g. The side play is measured a second time with the universal joint rotated 90 degrees from the first side play measurement

4.7.3 Axial load (see 3.5.3). The axial load for a particular type and size of universal joint shall meet the requirements of 3.5.3. The following details shall apply:

a. The axial load shall be applied in both tension and compression for 30 seconds with angle A, as shown in Figure 1, equal to 0°.
b. After application of the axial load, the universal joint shall be inspected for permanent deformation and for tightness in accordance with 4.7.4.

4.7.4 Tightness (see 3.5.4). Tightness for universal joints shall meet the requirements of 3.5.4. The following details shall apply:

a. Universal joints shall be tested for tightness with the lubricant retaining cover removed, and may be lubricated during the test.
b. The universal joint shall be mounted horizontally, and the axis of rotation shall be vertical for this test, such that the effects of gravity can be reduced/eliminated from the measurement (see Figure 2).
c. The moment of force required to move the upper end of the universal joint through the minimum angle of 30° shall be measured.

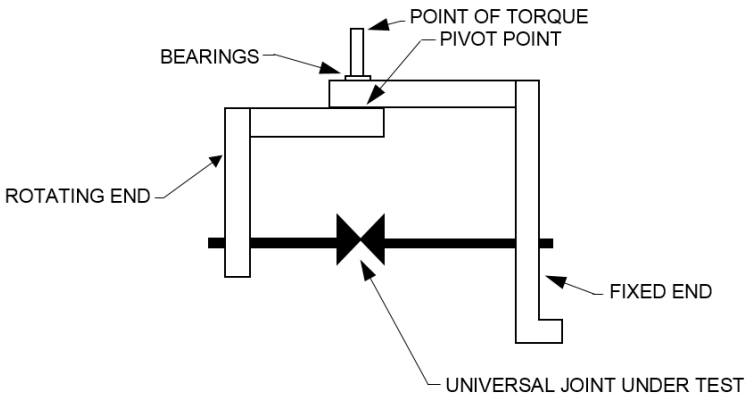


FIGURE 2. Example of test fixture for tightness.

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4.7.5 Static torque load (see 3.5.5). The universal joint shall meet the requirements of 3.5.5 when subjected to a static load test. The following details shall apply:

a. The universal joints shall be subjected to 90% of the static torque load specified in Table III, without exceeding 12° torsional deflection, and shall not fail.
b. If no failures occurred during the 90% static load test, the same universal joints shall then be subjected to the 100% torque load shown in Table III, and shall not fail.

4.7.6 Torsional play (see 3.5.6). Universal joints shall meet the requirements of 3.5.6 when subjected to torsional play. The following details shall apply:

a. With the lubrication retaining covers and lubricant removed, the applicable torque specified in Table III under “Torsional play” shall be applied to the universal joint to obtain an initial reading.
b. The torque load shall then be reversed and a second reading taken.
c. The torsional play at the circumference of the universal joint is the difference between the two readings.
    (1) The torsional play for samples that have not been subjected to endurance testing shall not exceed the limits specified in 3.5.6.
    (2) The torsional play for samples that have been subjected to endurance testing shall not exceed 5°.

4.7.7 Endurance (see 3.5.7). Universal joints shall meet the requirements of 3.5.7 when subjected to endurance testing. The following details shall apply:

a. Test specimens:
    (1) Class 1 universal joints shall be submerged in lubricating oil conforming to MIL-PRF-7870 before starting the endurance test.
    (2) Class 2 universal joints shall be subjected to the endurance test with lubricant retaining covers in place.

b. Test equipment shall be of such design and construction that the test specimens will be free from inadvertent end loads during the entire test. The following details shall apply:
    (1) A means shall be provided to avoid high starting torque loads.
    (2) The test equipment shall be a smooth-running mechanism free from vibration and shock loads.
    (3) After startup, the dynamic torque load shall be constant.

c. Testing:
    (1) Forced cooling shall be used during the test.
    (2) The universal joints shall be continuously operated at 120 rotation per minute under the applicable dynamic torque loads and for the lengths of runs shown in Table III, with the angle A shown in Figure 1 equal to 15°.

4.7.8 Lubricant retention (see 3.5.8). Universal joints shall meet the requirements of 3.5.8 when subjected to lubricant retention testing. The following details shall apply:

a. Test specimens shall be universal joints with lubrication covers that have successfully passed endurance testing specified in 4.7.7.
b. Test specimens shall be exposed to a temperature of -65°F (-53.89°C) for 24 hours.
c. The universal joints shall then be operated at angle A of 20° (see Figure 1) for 1 hour at 3 rotations per minute.
d. The temperature shall then be raised to 160°F (71.11°C) for 160 hours and the universal joint operated at an angle A of 20° (see Figure 1) for 1 hour at 3 rotations per minute.
e. Failure of the lubrication retaining cover shall also be considered failure of the universal joint.

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4.7.9 Ozone (see 3.4.5.1). Unless specified otherwise, all rubber or elastomeric components used on the universal joint shall meet the requirements of 3.4.5.1 when exposed to ozone. Rubber or elastomeric components shall be tested in accordance with ASTM D1149, Method B, Procedure B4-A, with a 70-hour exposure time.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Controls Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES
(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The universal joints covered by this specification are used or intermittent mechanical linkages, i.e. gun controls, steering, and flap/slat actuation on aerospace equipment. The universal joints described in this specification are military unique, as they must be able to operate satisfactorily when subjected to torsional and axial overload capacity, and low torsional deflection. Commercial universal joints are not designed to withstand these extreme conditions or sudden mechanical changes, and would experience catastrophic failures.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Applicable specification sheet number, title, and date.
c. Applicable specification sheet PIN (see 1.2).
d. Packaging requirements (see 5.1)
e. Shelf-life requirements (see 6.4).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No.6193 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, P.O. Box 3990, ATTN: DSCC-VQ, Columbus, Ohio 43218-3990 or emailed to vqp.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil..

6.3.1 Provisions governing qualification (SD-6). Copies of "Provisions Governing Qualification" are available online at https://assist.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.

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6.4 Shelf life. This specification covers items where the assignment of a Federal shelf-life code is a consideration. Specific shelf-life requirements should be specified in the contract or purchase order, and should include, as a minimum, shelf-life code, shelf-life package markings in accordance with MIL-STD-129 or FED-STD-123, preparation of a materiel quality storage standard for type II (extendible) shelf-life items, and a minimum of 85 percent shelf-life remaining at time of receipt by the Government. These and other requirements, if necessary, are in DoD4140.27-M, *Shelf- life Management Manual*. The shelf-life codes are in the Federal Logistics Information System Total Item Record. Additive information for shelf-life management may be obtained from DoD 4140.27-M, or the designated shelf-life Points of Contact (POC). The POC should be contacted in the following order: (1) the Inventory Control Points that manage the item and (2) the DoD Service and Agency administrators for the DoD Shelf-Life Program. Appropriate POCs for the DoD Shelf-Life Program can be contacted through the DoD Shelf-Life Management website: https://www.shelflife.hq.dla.mil/.

6.5 Supersession information. The MS specification sheets associated with MIL-J-6193C have been canceled, and are superseded by the specification sheets listed in 2.2.1.

6.5.1 Substitutability information. Items in inventory having a PIN identified in the canceled MS specification sheets may be substituted for the corresponding item identified by a PIN that has been assigned by the preparing activity under this specification. Existing items in inventory need not be re-identified with the PIN assigned to newly acquired items.

6.6 Guidance on the use of alternative parts with less hazardous or non-hazardous materials. This specification provides for a number of alternative plating materials via the PIN. Users should select the PIN with the least hazardous material that meets the form, fit, and function requirements of their application.

6.7 Subject term (key word) listing.

Cadmium
Nickel
Ozone
Plating
Zinc

6.8 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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CONCLUDING MATERIAL

Custodians:
Army – AR
Navy – AS
Air Force – 71

Preparing activity:
DLA – CC
(Project 3010-2025-001)

Review activities:
Army – AV
Navy – CG, MC, SA

Industry association:
SAE – A6

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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